



Islamic University Of Technology

**EEE 4401: Power System I
Complex Engineering Problem**

Submitted by:

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Section : C

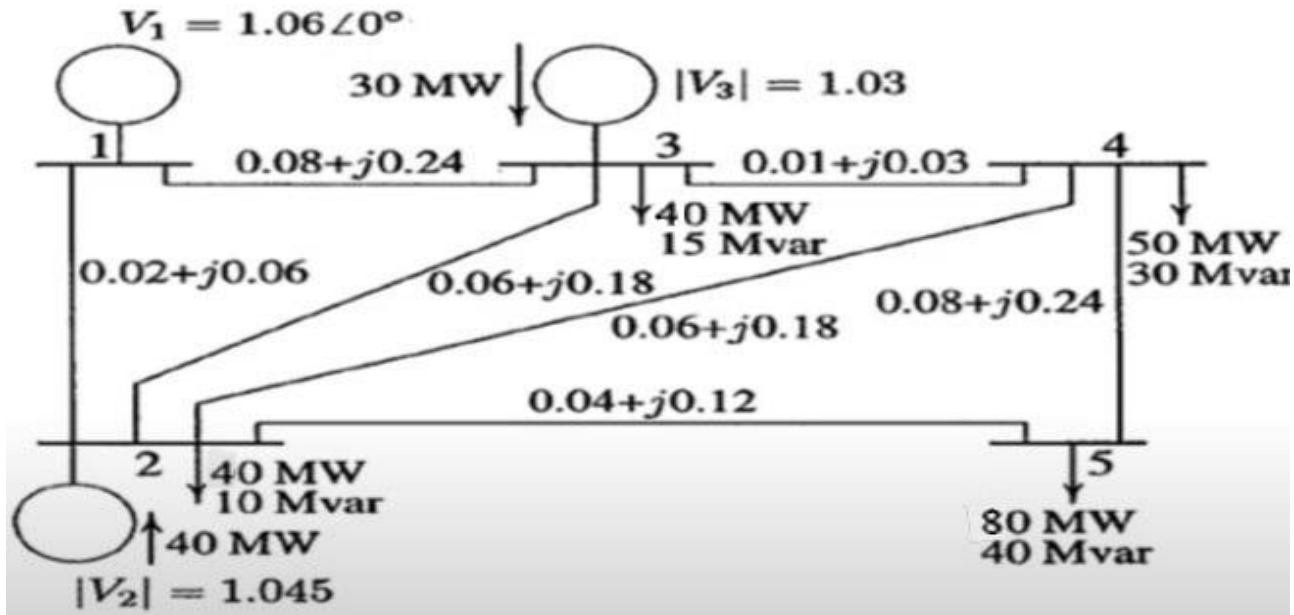
Topic : Power System for Load Flow Analysis

Submission : 07 May, 2024

Part A: Selection of Suitable Power System for Load Flow Analysis

Introduction:

Reviewing relevant research papers and books, I have selected the following Power System for analysis:



This System has 5 Buses:

Bus-1 : Slack Bus, Bus voltage is $1.06 \angle 0^\circ$ p.u.

Bus-2 : Generation Bus, Voltage magnitude is 1.045 p.u. and Real power supply is 40 MW

Bus-3 : Generation Bus, Voltage magnitude is 1.03 p.u. and Real power supply is 30 MW

Bus-4 : Load Bus, load is 50 MW and 30 MVar

Bus-5 : Load Bus, load is 80 MW and 40 MVar

Total Real power demand of this system, $P = 40+40+50+80$ MW = **210 MW**

For real power, it is clear that **$200 \leq P \leq 250$**

Total Reactive power demand of this system, $Q = 10+15+30+40$ MVar = **95 MVar**

For reactive power, it is clear that **$80 \leq Q \leq 100$**

To sum up, my selected power system meets all the required Specifications given in the question paper. So, this system can be used for further analysis.

Part B: Load Flow Analysis

Task-01: Load Flow Analysis using MATLAB

MATLAB code for generating Y_{Bus} :

```
Editor - E:\IUT\4th Semester\EEE 4402\Lab sheets\ybus.m
ybus.m  Graph_ploting.mlx  EEE4408_Lab_02.mlx  OLE_4408.mlx
1  function[Ybus] = ybus(zdata)
2  - nl=zdata(:,1); % starting BUS
3  - nr=zdata(:,2); % Ending BUS
4  - R=zdata(:,3); % branch resistance
5  - X=zdata(:,4); % branch reactance
6  - nbr=length(zdata(:,1)); % No of branches
7  - nbus = max(max(nl), max(nr)); % No of BUS
8  - Z = R + j*X; % Branch impedance
9  - y= ones(nbr,1)./Z; % Branch admittance
10 % initialize Ybus to zero
11 - Ybus=zeros(nbus,nbus);
12 % formation of the off diagonal elements
13 - for k = 1:nbr;
14 -     if nl(k) > 0 & nr(k) > 0
15 -         Ybus(nl(k),nr(k)) = Ybus(nl(k),nr(k)) - y(k);
16 -         Ybus(nr(k),nl(k)) = Ybus(nl(k),nr(k));
17 -     end
18 - end
19 % formation of the diagonal elements
20 - for n = 1:nbus
21 -     for k = 1:nbr
22 -         if nl(k) == n | nr(k) == n
23 -             Ybus(n,n) = Ybus(n,n) + y(k);
24 -         else, end
25 -     end
26 - end
```

Output (Y_{Bus} in rectangular form) :

```
Command Window
>> zdata=[1 2 0.02 0.06;1 3 0.08 0.24;2 3 0.06 0.18;2 4 0.06 0.18;2 5 0.04 0.12;3 4 0.01 0.03;4 5 0.08 0.24]

zdata =

    1.0000    2.0000    0.0200    0.0600
    1.0000    3.0000    0.0800    0.2400
    2.0000    3.0000    0.0600    0.1800
    2.0000    4.0000    0.0600    0.1800
    2.0000    5.0000    0.0400    0.1200
    3.0000    4.0000    0.0100    0.0300
    4.0000    5.0000    0.0800    0.2400

>> ybus(zdata)

ans =

    6.2500 -18.7500i   -5.0000 +15.0000i   -1.2500 + 3.7500i    0.0000 + 0.0000i    0.0000 + 0.0000i
   -5.0000 +15.0000i   10.8333 -32.5000i   -1.6667 + 5.0000i   -1.6667 + 5.0000i   -2.5000 + 7.5000i
   -1.2500 + 3.7500i   -1.6667 + 5.0000i   12.9167 -38.7500i  -10.0000 +30.0000i    0.0000 + 0.0000i
    0.0000 + 0.0000i   -1.6667 + 5.0000i  -10.0000 +30.0000i   12.9167 -38.7500i  -1.2500 + 3.7500i
    0.0000 + 0.0000i   -2.5000 + 7.5000i    0.0000 + 0.0000i   -1.2500 + 3.7500i    3.7500 -11.2500i

>> Ybus_rectangular=ans

Ybus_rectangular =

    6.2500 -18.7500i   -5.0000 +15.0000i   -1.2500 + 3.7500i    0.0000 + 0.0000i    0.0000 + 0.0000i
   -5.0000 +15.0000i   10.8333 -32.5000i   -1.6667 + 5.0000i   -1.6667 + 5.0000i   -2.5000 + 7.5000i
   -1.2500 + 3.7500i   -1.6667 + 5.0000i   12.9167 -38.7500i  -10.0000 +30.0000i    0.0000 + 0.0000i
    0.0000 + 0.0000i   -1.6667 + 5.0000i  -10.0000 +30.0000i   12.9167 -38.7500i  -1.2500 + 3.7500i
    0.0000 + 0.0000i   -2.5000 + 7.5000i    0.0000 + 0.0000i   -1.2500 + 3.7500i    3.7500 -11.2500i
```

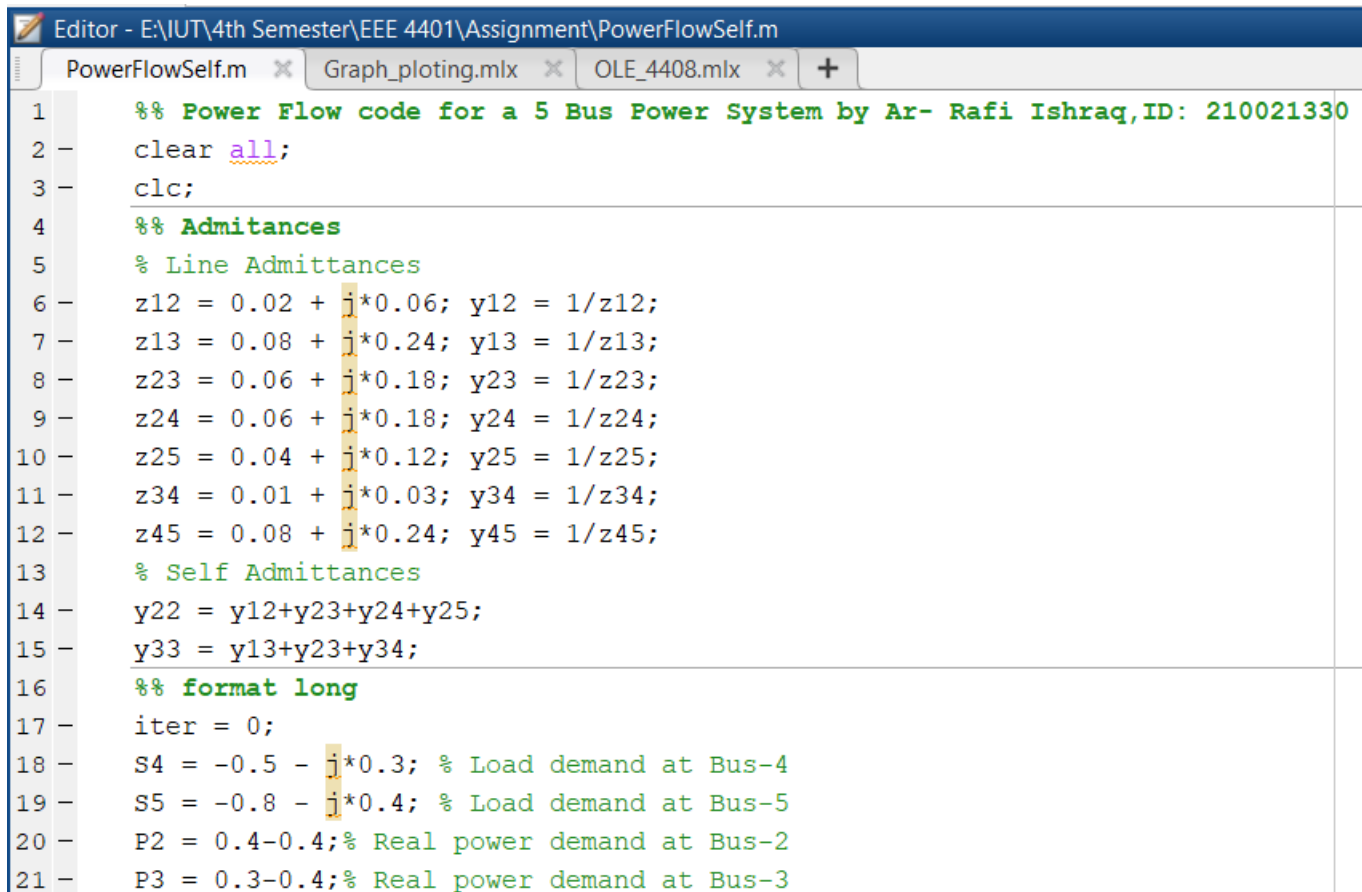
Output (Y_{Bus} in polar form) :

```
>> % Convert rectangular values to polar form
Ybus_magnitude = abs(Ybus_rectangular); % Magnitude
Ybus_angle = angle(Ybus_rectangular); % Angle in radians

% Display the polar form of the Ybus matrix
disp('Magnitude (S) of Ybus Matrix:');
disp(Ybus_magnitude);
disp('Angle (radians) of Ybus Matrix:');
disp(Ybus_angle);
Magnitude (S) of Ybus Matrix:
    19.7642    15.8114     3.9528         0         0
    15.8114    34.2580     5.2705     5.2705     7.9057
     3.9528     5.2705    40.8461    31.6228         0
         0     5.2705    31.6228    40.8461     3.9528
         0     7.9057         0     3.9528    11.8585

Angle (radians) of Ybus Matrix:
    -1.2490     1.8925     1.8925         0         0
     1.8925    -1.2490     1.8925     1.8925     1.8925
     1.8925     1.8925    -1.2490     1.8925         0
         0     1.8925     1.8925    -1.2490     1.8925
         0     1.8925         0     1.8925    -1.2490
```

MATLAB code for power flow analysis:



The screenshot shows the MATLAB Editor window with the file 'PowerFlowSelf.m' open. The script is titled '% Power Flow code for a 5 Bus Power System by Ar- Rafi Ishraq, ID: 210021330'. The code includes comments for 'Admittances' and 'Self Admittances', and defines variables for line impedances, admittances, and load demands at various buses.

```
Editor - E:\JUT\4th Semester\EEE 4401\Assignment\PowerFlowSelf.m
PowerFlowSelf.m  Graph_ploting.mlx  OLE_4408.mlx  +
1  %% Power Flow code for a 5 Bus Power System by Ar- Rafi Ishraq, ID: 210021330
2  clear all;
3  clc;
4  %% Admittances
5  % Line Admittances
6  z12 = 0.02 + j*0.06; y12 = 1/z12;
7  z13 = 0.08 + j*0.24; y13 = 1/z13;
8  z23 = 0.06 + j*0.18; y23 = 1/z23;
9  z24 = 0.06 + j*0.18; y24 = 1/z24;
10 z25 = 0.04 + j*0.12; y25 = 1/z25;
11 z34 = 0.01 + j*0.03; y34 = 1/z34;
12 z45 = 0.08 + j*0.24; y45 = 1/z45;
13 % Self Admittances
14 y22 = y12+y23+y24+y25;
15 y33 = y13+y23+y34;
16 %% format long
17 iter = 0;
18 S4 = -0.5 - j*0.3; % Load demand at Bus-4
19 S5 = -0.8 - j*0.4; % Load demand at Bus-5
20 P2 = 0.4-0.4j; % Real power demand at Bus-2
21 P3 = 0.3-0.4j; % Real power demand at Bus-3
```

```

22 % Initial Bus Voltage Values
23 - V1 = 1.06 + j*0; % Constant Voltage
24 - V2 = 1.045 + j*0; % Real part is constant
25 - V3 = 1.03 + j*0; % Real part is constant
26 - V4 = 1 + j*0;
27 - V5 = 1 + j*0;
28 - Vm2 = 1.045;
29 - Vm3 = 1.03;

```

```

30 %% Iteration Loop for Gauss-Seidel Method

```

```

31 - for I = 1:10 % Loop will do 10 Iterations
32 -     iter = iter+1; % No. of iteration
33 -     E2=V2;
34 -     E3=V3;
35 -     E4=V4;
36 -     E5=V5;
37     % Determining Voltage at Bus-5
38 -     V5 = ((conj(S5)/conj(V5))+y25*V2+y45*V4)/(y25+y45);
39 -     DV5=V5-E5;
40     % Determining Voltage at Bus-2
41 -     Q2 = -imag(conj(V2)*(y22*V2-y12*V1-y23*V3-y24*V4-y25*V5));
42 -     S2 = P2 + j*(Q2-0.1);
43 -     Vc2 = ((conj(S2)/conj(V2))+y12*V1+y23*V3+y24*V4+y25*V5)/(y22);
44 -     Vi2 = imag(Vc2);
45 -     Vr2 = sqrt(Vm2^2 - Vi2^2);
46 -     V2 = Vr2 + j*Vi2;
47 -     DV2 = V2-E2;
48     % Determining Voltage at Bus-4
49 -     V4 = ((conj(S4)/conj(V4))+y24*V2+y34*V3+y45*V5)/(y24+y34+y45);
50 -     DV4 = V4 - E4;
51     % Determining Voltage at Bus-3
52 -     Q3 = -imag(conj(V3)*(y33*V3-y13*V1-y23*V2-y34*V4));
53 -     S3 = P3 + j*(Q3-0.15);
54 -     Vc3 = ((conj(S3)/conj(V3))+y13*V1+y23*V2+y34*V4)/(y33);
55 -     Vi3 = imag(Vc3);
56 -     Vr3 = sqrt(Vm3^2 - Vi3^2);
57 -     V3 = Vr3 + j*Vi3;
58 -     DV3 = V3-E3;
59 - end

```

```

60 %% Calculation of Currents

```

```

61 - I12 = y12*(V1-V2); I21 = -I12;
62 - I13 = y13*(V1-V3); I31 = -I13;
63 - I23 = y23*(V2-V3); I32 = -I23;
64 - I24 = y24*(V2-V4); I42 = -I24;
65 - I25 = y25*(V2-V5); I52 = -I25;
66 - I34 = y34*(V3-V4); I43 = -I34;
67 - I45 = y45*(V4-V5); I54 = -I45;

```

```

68 %% Calculation of Complex Powers
69 - S12=(V1*conj(I12))*100; S21=(V2*conj(I21))*100;
70 - S13=(V1*conj(I13))*100; S31=(V3*conj(I31))*100;
71 - S23=(V2*conj(I23))*100; S32=(V3*conj(I32))*100;
72 - S24=(V2*conj(I24))*100; S42=(V4*conj(I42))*100;
73 - S25=(V2*conj(I25))*100; S52=(V5*conj(I52))*100;
74 - S34=(V3*conj(I34))*100; S43=(V4*conj(I43))*100;
75 - S45=(V1*conj(I45))*100; S54=(V2*conj(I54))*100;

76 %% Calculation of Power Loss
77 - SL12=S12+S21;
78 - SL13=S13+S31;
79 - SL23=S23+S32;
80 - SL24=S24+S42;
81 - SL25=S25+S52;
82 - SL34=S34+S43;
83 - SL45=S45+S54;

84 %% Display Output
85 - disp('Bus Voltages in p.u. : ')
86 - V1 == abs(V1),V2 == abs(V2),V3 == abs(V3),V4 == abs(V4),V5 == abs(V5)
87 - disp('Power Losses: (units: MW & MVar) ')
88 - SL12,SL13,SL23,SL24,SL25,SL34,SL45
89 - disp('Total Real Power Loss: (MW) ')
90 - Ptotal == real(SL12+SL13+SL23+SL24+SL25+SL34+SL45)
91 - disp('Total Reactive Power Loss: (MVar) ')
92 - Qtotal == imag(SL12+SL13+SL23+SL24+SL25+SL34+SL45)

```

Output of Power Flow:

Command Window

Bus Voltages in p.u. :

V1 =

1.0600

V2 =

1.0450

V3 =

1.0300

V4 =

1.0163

V5 =

0.9796

Command Window

Power Losses: (units: MW & MVar)

SL12 =

1.4634 + 4.3903i

SL13 =

0.7476 + 2.2429i

SL23 =

0.1012 + 0.3035i

SL24 =

0.2717 + 0.8151i

SL25 =

1.9930 + 5.9790i

SL34 =

0.2676 + 0.8028i

SL45 =

-0.3218 + 0.9970i

Total Real Power Loss: (MW)

Ptotal =

4.5228

Total Reactive Power Loss: (MVar)

Qtotal =

15.5307

Result From Book:

Bus No.	Voltage		Angle Degree	-----Load-----		--Generation--		Inj
	Mag.			MW	Mvar	MW	Mvar	
1	1.060		0.000	0.000	0.00	83.051	7.271	
2	1.045		-1.782	20.000	10.00	40.000	41.811	
3	1.030		-2.664	20.000	15.00	30.000	24.148	
4	1.019		-3.243	50.000	30.00	0.000	0.000	
5	0.990		-4.405	60.000	40.00	0.000	0.000	
Total				150.000	95.000	153.051	73.230	

Comparison Table:

	Result of Matlab	Solution of Book
V1	1.06 p.u.	1.06 p.u.
V2	1.045 p.u.	1.045 p.u.
V3	1.03 p.u.	1.03 p.u.
V4	1.01 p.u.	1.019 p.u.
V5	0.9796 p.u.	0.98 p.u.
Real Power Loss	4.5228 MW	4.62 MW
Reactive Power Loss	15.5307 MVar	15.54 MVar

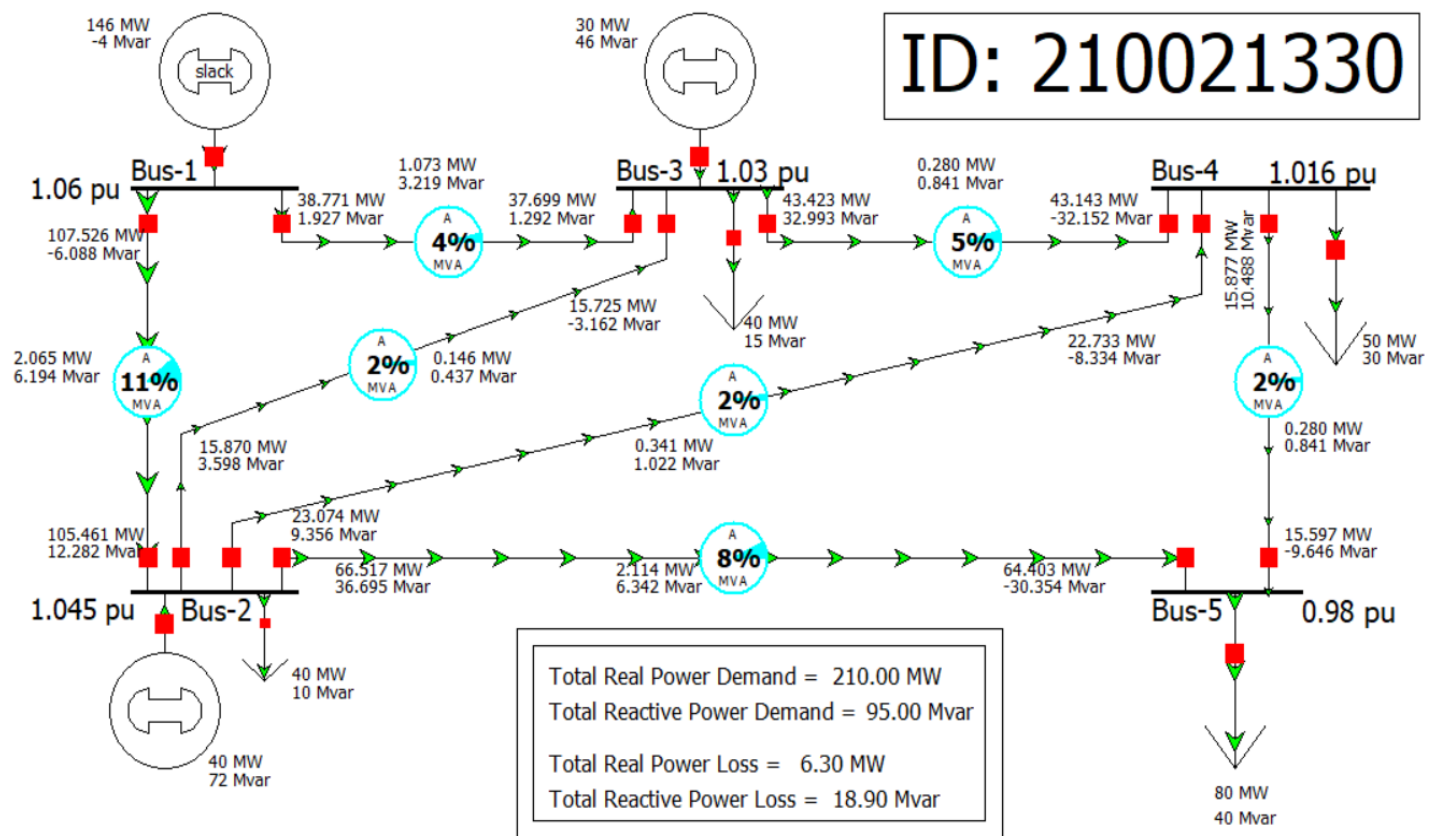
Comments:

The MATLAB-based Gauss-Seidel load-flow analysis produced a stable voltage profile for the selected 5-bus system. The voltage magnitudes at the slack and generation buses remained at their specified values, while the load-bus voltages changed according to the connected loads and network impedances. Bus 5 experienced the lowest voltage magnitude, which is consistent with its comparatively large load demand. The calculated active and reactive power losses indicate the effect of transmission-line resistance and reactance on system performance. Overall, the obtained results demonstrate successful application of the Gauss-Seidel method for load-flow analysis.

Task-02: Load Flow Analysis using Power World Simulator

Simulation of the selected System:

ID: 210021330



Comparison Table:

	Result of Power World Simulator	Solution of Book
V1	1.06 p.u.	1.06 p.u.
V2	1.045 p.u.	1.045 p.u.
V3	1.03 p.u.	1.03 p.u.
V4	1.016 p.u.	1.019 p.u.
V5	0.98 p.u.	0.98 p.u.
Real Power Loss	6.03 MW	6.028 MW
Reactive Power Loss	18.90 MVar	18.902 MVar

Comments:

The PowerWorld simulation shows the active and reactive power flow across the 5-bus system. The slack and generator buses maintain their specified voltages, while the load-bus voltages vary with network loading. Bus 5 has the lowest voltage due to its higher load demand. The simulation also illustrates transmission losses and provides graphical verification of the MATLAB load-flow results.

Power Loss Minimization

For reducing the power loss in the system, I have made the line resistance value $1/100^{\text{th}}$ of its previous value and the line reactance value $1/10^{\text{th}}$ of its previous value. This change in the line impedance has reduced the power loss in the system. Specially, Real power loss is reduced significantly.

Updated code part for minimizing power loss:

```
Editor - E:\IUT\4th Semester\EEE 4401\Assignment\PowerFlowSelf.m
PowerFlowSelf.m  Graph_plotting.mlx  OLE_4408.mlx  +
1  %% Power Flow code for a 5 Bus Power System by Ar- Rafi Ishraq, ID: 210021330
2  clear all;
3  clc;
4  %% Admittances
5  % Line Admittances
6  z12 = 0.0002 + j*0.006; y12 = 1/z12;
7  z13 = 0.0008 + j*0.024; y13 = 1/z13;
8  z23 = 0.0006 + j*0.018; y23 = 1/z23;
9  z24 = 0.0006 + j*0.018; y24 = 1/z24;
10 z25 = 0.0004 + j*0.012; y25 = 1/z25;
11 z34 = 0.0001 + j*0.003; y34 = 1/z34;
12 z45 = 0.0008 + j*0.024; y45 = 1/z45;
13 % Self Admittances
14 y22 = y12+y23+y24+y25;
15 y33 = y13+y23+y34;
16 %% format long
17 iter = 0;
18 S4 = -0.5 - j*0.3; % Load demand at Bus-4
19 S5 = -0.8 - j*0.4; % Load demand at Bus-5
20 P2 = 0.4-0.4j; % Real power demand at Bus-2
21 P3 = 0.3-0.4j; % Real power demand at Bus-3
22 % Initial Bus Voltage Values
23 V1 = 1.06 + j*0; % Constant Voltage
24 V2 = 1.045 + j*0; % Real part is constant
25 V3 = 1.03 + j*0; % Real part is constant
26 V4 = 1 + j*0;
27 V5 = 1 + j*0;
28 Vm2 = 1.045;
29 Vm3 = 1.03;
```

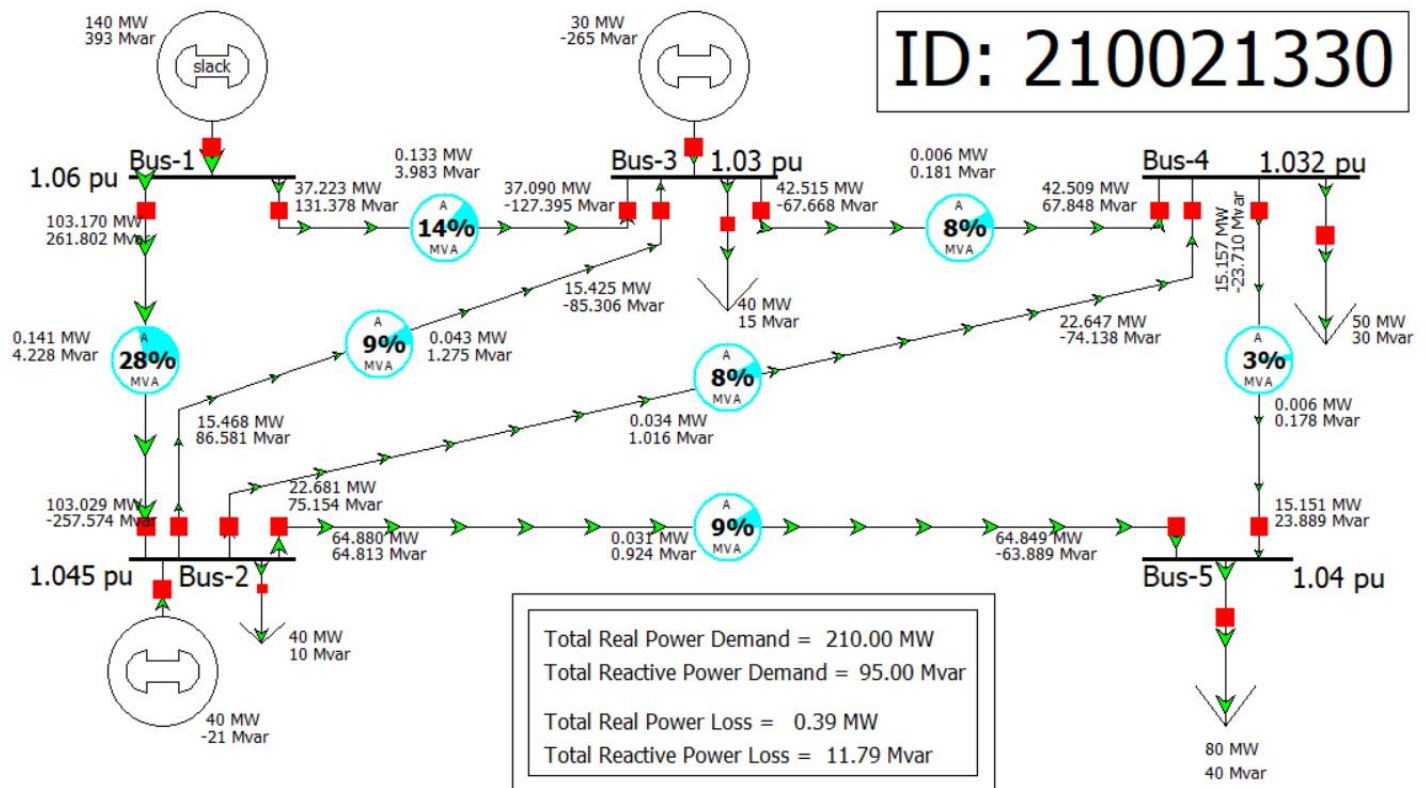
Updated Power Flow:

Command Window	Command Window	
Bus Voltages in p.u. :	Power Losses: (units: MW & MVar)	
V1 =	SL12 =	
1.0600	0.1384 + 4.1526i	
V2 =	SL13 =	
1.0450	0.1314 + 3.9433i	
V3 =	SL23 =	
1.0300	0.0423 + 1.2701i	SL45 =
V4 =	SL24 =	0.3400 - 0.2689i
1.0319	0.0335 + 1.0039i	Total Real Power Loss: (MW)
V5 =	SL25 =	Ptotal =
1.0373	0.0298 + 0.8947i	0.7212
	SL34 =	Total Reactive Power Loss: (MVar)
	0.0057 + 0.1721i	Qtotal =
		11.1677

Comments:

After reducing the transmission-line resistance and reactance values, a significant reduction in system power loss was observed. The lower line impedance reduced the voltage drop across the transmission network and consequently improved the voltage magnitudes at the load buses. In particular, the voltages of Bus 4 and Bus 5 increased compared with the original system. The active power loss decreased substantially, while the reactive power loss was also reduced. This result demonstrates the strong relationship between transmission-line impedance, voltage regulation, and network power losses. However, the modified line parameters represent a theoretical sensitivity study rather than a practical transmission-line optimization procedure.

Updated Simulation:



Conclusion:

This project investigated the load-flow characteristics of a 5-bus power system using both MATLAB and PowerWorld Simulator. A suitable network satisfying the specified real and reactive power-demand requirements was selected, and its bus-admittance matrix was determined. The Gauss-Seidel method was then implemented in MATLAB to calculate bus voltages, branch power flows, and transmission losses. The same network was modeled in PowerWorld Simulator to visualize active and reactive power flows and assess the system performance using an independent simulation environment. The results showed that bus voltages and network losses are strongly influenced by transmission-line parameters and load distribution. A further sensitivity study using reduced line impedance demonstrated substantial reduction in transmission losses and improvement in load-bus voltage magnitudes. Overall, the project provided practical experience in power-system modeling, numerical load-flow analysis, simulation, and interpretation of power-system operating conditions.